

When Marginal Shift Magnitudes Cannot Identify Conformal Coverage Loss

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Can the magnitude of distribution shift, or the distribution of model confidence, identify how much conformal coverage is lost?

CENTRAL IDEA

Separate drift detection from coverage identification. First analyze a known binary logistic label law; then replace that strong assumption with calibration, target confidence frequencies and a density-ratio bound.

MAIN FINDING

Rotation-invariant discrepancies can agree on Gaussian shifts with coverage gaps approaching the sharp bound $1/2 - \alpha$. Under the known link, the complete coverage curve is equivalent to the absolute-score distribution. Under weaker information, a sharp partial-identification envelope replaces a point answer.

EVIDENCE AND SCOPE

The same two target laws attain the envelope endpoints simultaneously at every threshold. A finite-sample version uses source labels and unlabeled target scores over a prespecified threshold grid. Simulations, independent optimization checks and learned-score replay on two UCI datasets illustrate the distinct regimes.

LIMITS

Perfect source calibration is insufficient in general. Confidence suffices only when conditional errors have no hidden variation within confidence levels, apart from degenerate weight constraints. The different guarantees require different assumptions.

TAKEAWAY

A drift alarm measures change; a coverage statement needs information about how that change interacts with errors.

한국어 핵심 요약

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